

MATH 375: Probability Theory

Mariana Restrepo

Fall 2026

Lecture 6

Contents

1	Lecture 6: Independence and Repeated Trials	1
1.1	Independence and complements	1
1.2	Independence of three events	2
1.3	Repeated independent trials	2
1.4	Which dice sum appears first?	2
1.5	Three distinguishable dice	3
1.6	Example: rain and lateness	3
1.7	Example: selecting a coin	3

1 Lecture 6: Independence and Repeated Trials

1.1 Independence and complements

Events E and F are independent when

$$P(E \cap F) = P(E)P(F).$$

If E and F are independent, then E and F^c are also independent. To see why, split E into disjoint pieces:

$$E = (E \cap F) \cup (E \cap F^c).$$

Consequently,

$$\begin{aligned} P(E \cap F^c) &= P(E) - P(E \cap F) \\ &= P(E) - P(E)P(F) \\ &= P(E)(1 - P(F)) \\ &= P(E)P(F^c). \end{aligned}$$

This is exactly the condition for E and F^c to be independent.

1.2 Independence of three events

Three events E , F , and G are **mutually independent** when all four conditions hold:

$$\begin{aligned}P(E \cap F) &= P(E)P(F), \\P(E \cap G) &= P(E)P(G), \\P(F \cap G) &= P(F)P(G), \\P(E \cap F \cap G) &= P(E)P(F)P(G).\end{aligned}$$

Checking only the three pairs is not enough to establish mutual independence.

1.3 Repeated independent trials

Suppose each trial independently succeeds with probability p . Then each trial fails with probability $1 - p$.

At least one success in n trials. The complement is “all n trials fail,” with probability $(1 - p)^n$. Therefore,

$$P(\text{at least one success}) = 1 - (1 - p)^n.$$

Exactly k successes in n trials. First choose which k positions contain successes. For any particular choice of positions, independence gives probability $p^k(1 - p)^{n-k}$. Hence

$$P(\text{exactly } k \text{ successes}) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

1.4 Which dice sum appears first?

Independently roll two fair dice until either a sum of 5 or a sum of 7 appears. What is the probability that 5 appears first?

In one roll,

$$P(\text{sum } 5) = \frac{4}{36}, \quad P(\text{sum } 7) = \frac{6}{36}.$$

The probability of getting neither is

$$1 - \frac{4}{36} - \frac{6}{36} = \frac{26}{36}.$$

For 5 to appear for the first time on roll n before any 7, the first $n - 1$ rolls must be neither 5 nor 7, followed by a 5:

$$P(\text{first qualifying result is 5 on roll } n) = \left(\frac{26}{36}\right)^{n-1} \frac{4}{36}.$$

Add over all possible values of n :

$$\begin{aligned}P(5 \text{ before } 7) &= \sum_{n=1}^{\infty} \left(\frac{26}{36}\right)^{n-1} \frac{4}{36} \\&= \frac{4/36}{1 - 26/36} \\&= \boxed{\frac{2}{5}}.\end{aligned}$$

1.5 Three distinguishable dice

Roll a blue die B , a yellow die Y , and a red die R . What is the probability that

$$B < Y < R?$$

For the values to be strictly increasing, they must first be *distinct*. Choose any three of the six faces. Once chosen, exactly one arrangement places them in the order $B < Y < R$:

$$\#\{\text{favorable outcomes}\} = \binom{6}{3} = 20.$$

There are $6^3 = 216$ possible ordered outcomes, so

$$P(B < Y < R) = \frac{\binom{6}{3}}{6^3} = \frac{20}{216} = \frac{5}{54}.$$

1.6 Example: rain and lateness

Suppose it rains tomorrow with probability 0.7. Joe is late with probability 0.3 on rainy days and 0.1 on nonrainy days.

Let R mean “rain” and L mean “Joe is late.” By the law of total probability,

$$\begin{aligned} P(L) &= P(R)P(L | R) + P(R^c)P(L | R^c) \\ &= (0.7)(0.3) + (0.3)(0.1) \\ &= \boxed{0.24}. \end{aligned}$$

Now suppose Joe was *early*, meaning he was not late. What is the probability that it rained? First,

$$P(L^c) = 1 - 0.24 = 0.76, \quad P(L^c | R) = 1 - 0.3 = 0.7.$$

Bayes’ formula gives

$$P(R | L^c) = \frac{P(R)P(L^c | R)}{P(L^c)} = \frac{(0.7)(0.7)}{0.76} = \frac{49}{76}.$$

1.7 Example: selecting a coin

A box holds three coins: a two-headed coin, a fair coin, and a biased coin that lands heads with probability $\frac{3}{4}$. One coin is chosen uniformly and flipped. Given that the result is heads, what is the probability that the two-headed coin was chosen?

Let T , F , and D denote the three coin types, and let H mean “heads.” Each coin is chosen with probability $\frac{1}{3}$. The law of total probability gives

$$\begin{aligned} P(H) &= P(T)P(H | T) + P(F)P(H | F) + P(D)P(H | D) \\ &= \frac{1}{3}(1) + \frac{1}{3}\left(\frac{1}{2}\right) + \frac{1}{3}\left(\frac{3}{4}\right) \\ &= \frac{3}{4}. \end{aligned}$$

Therefore,

$$P(T | H) = \frac{P(T)P(H | T)}{P(H)} = \frac{(1/3)(1)}{3/4} = \frac{4}{9}.$$